

Can Monetary Policy Personas Influence LLM Interest Rate Decisions?

Li Yihe Sun Langda Li Bin Xia Tian Yang Huanyu

Track 2 Team 2 August 6, 2026

Abstract

This study examines whether predefined FOMC personas can systematically shift LLM monetary policy decisions when macroeconomic data remains unchanged. A factorial survey experiment is implemented on DeepSeek-V4-flash with fixed decoding parameters, covering three scenarios, five persona conditions and 15 repeated API calls per group. Persona instructions drive predictable policy divergence, with the Hawk – Dove rate spread reaching a maximum of 48.33 bps under mixed economic signals and an unbroken $\text{Hawk} \geq \text{Centrist} \geq \text{Dove}$ hierarchy in all cases. Such effects are limited to controlled prompt environments; the unprompted model baseline leans hawkish, and cross-model tests confirm directional but not identical persona responses. The results demonstrate that LLMs act as clean simulation tools to disentangle policymaker preference differences, removing human interference present in conventional economic research.

1. Introduction

Can explicitly assigned policy personas systematically alter a Large Language Model’s (LLM) interest-rate decisions when macroeconomic conditions are held strictly identical? Modern central banking relies not only on objective macroeconomic fundamentals but also on policymakers’ distinct interpretations of competing objectives—traditionally framed around hawkish and dovish monetary regimes. Because LLMs are pre-trained on vast corpora of financial commentary and economic literature, they provide a novel computational laboratory to simulate policymaker heterogeneity, enabling researchers to parse the net effects of policy priorities without human confounding factors.

Utilizing a standardized Factorial Survey Experiment (FSE) across three distinct macroeconomic scenarios, we evaluate whether role-based persona prompting generates theoretically coherent monetary policy differentiation. Our results yield two primary findings. First, persona framing exerts a powerful, directionally consistent net effect on rate decisions, strictly preserving the monotonic ordering of $\text{Hawk} \geq \text{Centrist} \geq \text{Dove}$ across all environments without any ranking reversals. Second, mixed-signal economic environments act as a "policy amplifier." When inflation and employment indicators send conflicting signals, the Hawk–Dove policy spread reaches a maximum of 48.33 basis points, indicating that persona variance is largest under ambiguous macroeconomic conditions where policy trade-offs are most severe. Third, persona prompts function as contextual interpretive frames that dictate attention allocation. Under mixed signals, they cleanly divide attention (e.g., hawks focus entirely on inflation), but under extreme

crises (unfavorable alignment), competing directives emerge and disperse the model's concerns. This explains why extreme macroeconomic conditions ultimately constrain persona-driven divergence.

2. Related work/Literature Review

This section contextualizes our experiment within five streams of literature that ground our findings, controls, and robustness checks.

LLM Role Simulation and Monetary Policy Heterogeneity (Supporting Result 5.1). LLMs function as non-deterministic simulators capable of generating persona-consistent behavior via role-based instructions (Shanahan et al., 2023; Argyle et al., 2023). In monetary economics, Istrefi (2019) shows that financial markets systematically categorize policymakers into inflation-focusing "hawks" and employment-focusing "doves". Together, this literature supports Result 5.1: persona prompts computationally reproduce directionally consistent Hawk–Dove policy heterogeneity across model environments.

Objective Trade-offs and Indicator Weight Allocation (Supporting Result 5.2). Central banking requires balancing competing objectives (Blinder, 1998), with interest rate decisions reflecting the relative weights assigned to inflation versus employment (Taylor, 1993). Crucially, policy trade-offs become most severe when macroeconomic indicators send conflicting signals. This explains why mixed-signal environments act as a "policy amplifier" that maximizes Hawk–Dove divergence.

Interpretive Frames and Competing Directives (Supporting Result 5.3). Persona prompts function as *contextual interpretive frames*, making specific policy priorities more salient during token generation (Lutz et al., 2025). However, when an extreme scenario's dominant empirical risk conflicts with a persona, it creates "unfavorable alignment" and competing directives that degrade behavioral stability (Zhang et al., 2025). This supports Result 5.3: interpretive frames cleanly guide attention under mixed signals, whereas extreme crises override personas and constrain policy divergence.

Default Orientations and Conditional Baseline Tendencies (Supporting Result 5.4). Unconditioned LLM outputs reflect systematic, model-specific opinion distributions rather than a neutral average (Santurkar et al., 2023). Pre-training corpora heavily emphasize central-bank inflation credibility, explaining why our unprompted baseline (No-Persona) exhibits a conditional hawkish tendency. While robustness checks confirm that this unprompted default is sensitive to model architecture and information density, explicit monetary policy personas exert a resilient, directionally stable effect.

Role Framing and Placebo Moderation Effects (Supporting Result 5.5). Generic role-play instructions alter reasoning patterns and task framing even without domain-specific guidance (Kong et al., 2024; Shanahan et al., 2023). This explains why an irrelevant academic Placebo

control moderates the model's default hawkish orientation and shifts attention toward a balanced concern under ambiguity. Consequently, using both unprompted and placebo controls is methodologically essential to isolate pure monetary policy preferences.

3. Method

The experiment uses a fully crossed synthetic-scenario design to isolate the causal effect of persona framing on LLM monetary-policy recommendations. All prompts, constraints, and output-validation rules were held constant except for the assigned persona and the numerical macroeconomic scenario.

3.1 Experimental design

The main experiment crosses five persona conditions (hawk, dove, centrist, no persona, and placebo) with three synthetic macroeconomic scenarios and 15 independent calls per cell, producing 225 observations. The three policy personas encode conventional differences in the relative priority assigned to price stability and maximum employment. The no-persona condition contains no policy role, while the placebo condition assigns an unrelated academic-researcher role without monetary-policy expertise or convictions. The scenarios represent high inflation, high unemployment, and mixed signals; each uses the same initial federal funds rate of 5.00% and the same six numerical indicators.

Across calls, the model, API endpoint, temperature (0.5), response format, rate-choice grid, prompt wording, and number of repetitions were fixed. Only the persona definition and scenario values varied. Calls were executed through the same pipeline, and each response was checked for valid JSON and for a `rate_decision_bps` value in $\{-50, -25, 0, +25, +50\}$.

3.2 Prompt and output schema

The system prompt instructs the model to act as the Federal Reserve Chair and to set the target federal funds rate for the upcoming quarter. For the hawk, dove, and centrist conditions, a persona-specific decision rule is inserted into the same prompt template. The no-persona condition omits a persona definition, whereas the placebo condition inserts a substantively irrelevant academic-researcher identity. The prompt requires the model to use the supplied numerical evidence, choose exactly one adjustment from the five-value grid, and return no prose outside a single JSON object. Full prompt text is reproduced in Appendix A.

The enforced schema contains five fields: `rate_decision_bps`, `target_rate_level`, `primary_concern`, `rationale`, and `confidence`. The `primary_concern` field is restricted to inflation, unemployment, financial stability, or balanced; `confidence` must lie between 0 and 1; and the `rationale` must contain two to four sentences.

3.3 Models

We conduct the main experiment using DeepSeek-V4-flash, accessed via DeepSeek's OpenAI-compatible API endpoint (api.deepseek.com/v1). For robustness checks across different models, we employ Qwen3.7-Plus as an alternative, which is accessed through Alibaba Cloud's DashScope API endpoint (dashscope.aliyuncs.com/api/v1), with the same prompt and parameter settings to ensure comparability.

4. Data

All non-LLM inputs are synthetic macroeconomic scenarios constructed by the authors. Each scenario consists of six numerical indicators presented without qualitative labels — scenarios are identified only as "Scenario A," "Scenario B," and "Scenario C" to avoid priming the model toward any particular interpretation of the data.

Table 1. Synthetic macroeconomic scenarios

Variable	Scenario A	Scenario B	Scenario C
Fed funds rate	5.0%	5.0%	5.0%
PCE inflation (YoY)	3.8%	2.1%	3.1%
Unemployment rate	3.8%	6.0%	4.9%
5-year inflation expectations	2.7%	2.1%	2.4%
Real GDP index	101.00	98.50	99.50
Potential GDP index	100.00	100.00	100.00

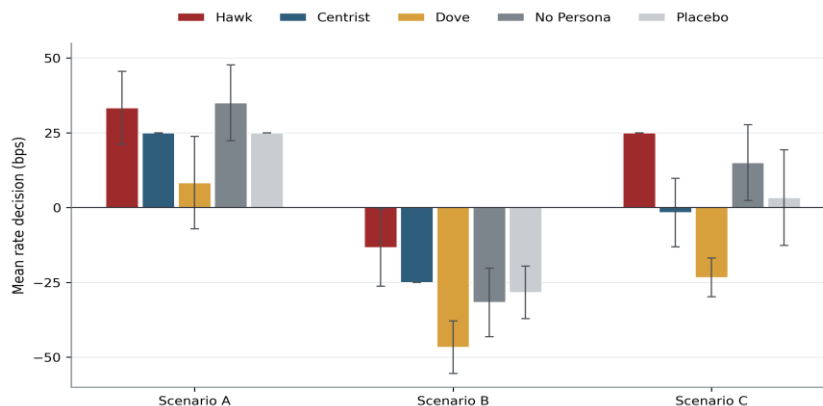


Figure 1: Mean rate decision (bps) by experimental group and scenario

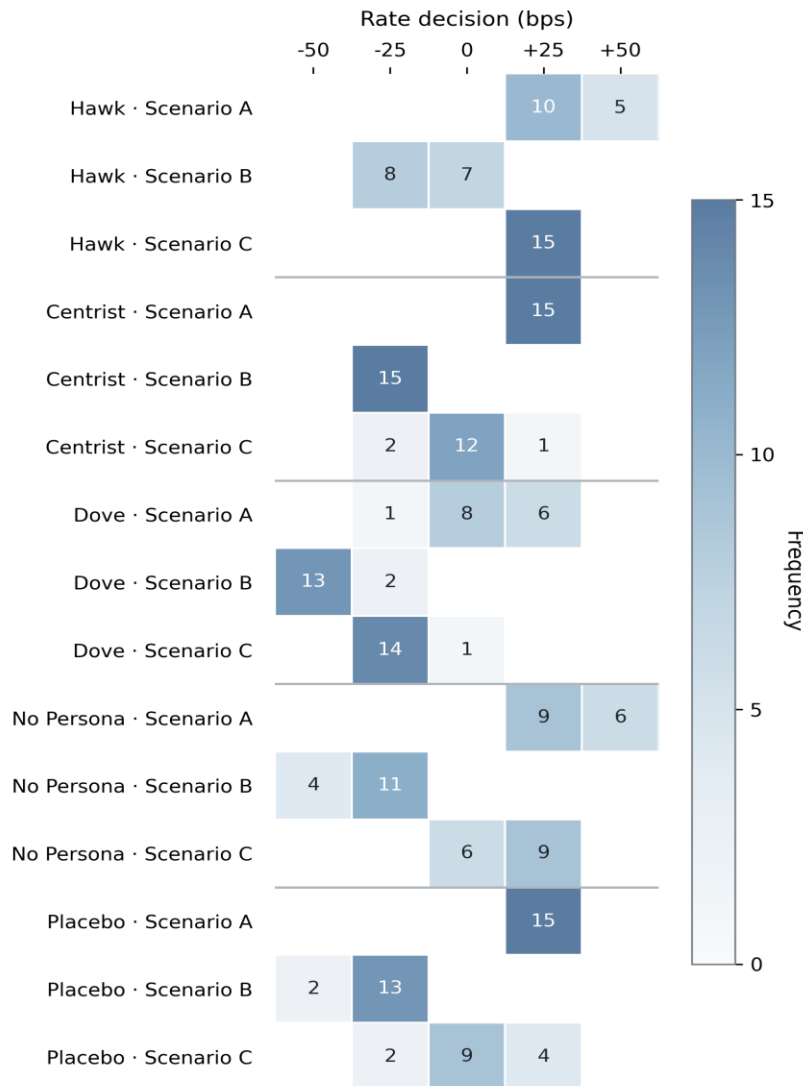


Figure 2: Frequency distribution of rate decisions across 15 repeated calls per cell

The three scenarios span distinct policy environments: Scenario A features elevated inflation (3.8%) with a tight labor market (unemployment at 3.8%) and GDP above potential — consistent with an overheating economy. Scenario B depicts a slack economy with unemployment at 6.0% and GDP below potential, while inflation sits at target. Scenario C presents mixed signals with moderately elevated inflation (3.1%) and unemployment (4.9%), where neither objective clearly dominates.

The model receives these six numbers through a standardized user prompt (reproduced in full in Appendix A) and must return a single JSON object with the following schema:

```
{
  "rate_decision_bps": <-50, -25, 0, +25, or +50>,

```

```

"target_rate_level": <float>,
"primary_concern": <"inflation" | "unemployment" | "financial_stability" | "balanced">,
"rationale": <2–4 sentence explanation>,
"confidence": <float in [0, 1]>
}

```

The `rate_decision_bps` field is hard-constrained to five discrete values representing the change in basis points from the current 5.00% federal funds rate. All other response fields are generated freely by the model.

Data were collected in a fully crossed design: 3 experimental groups (policy_persona, no_persona, placebo) $\times$ 3 scenarios $\times$ 15 repeated calls, yielding 225 observations. The policy_persona group contains 3 persona conditions (hawk, dove, centrist) while the no_persona and placebo groups each contain 1 condition, for a total of 15 unique group–persona–scenario cells. No observations were discarded; all 225 responses parsed successfully and were retained for analysis. Raw outputs are preserved in CSV format.

5. Results

5.1 Persona Prompting Effectively Differentiates Monetary Policy Perspectives in LLM Decisions

The results demonstrate clear differentiation across monetary policy personas. Under identical macroeconomic scenarios, hawkish, centrist, and dovish personas consistently generate different interest rate recommendations, following the expected ordering across all three scenarios:

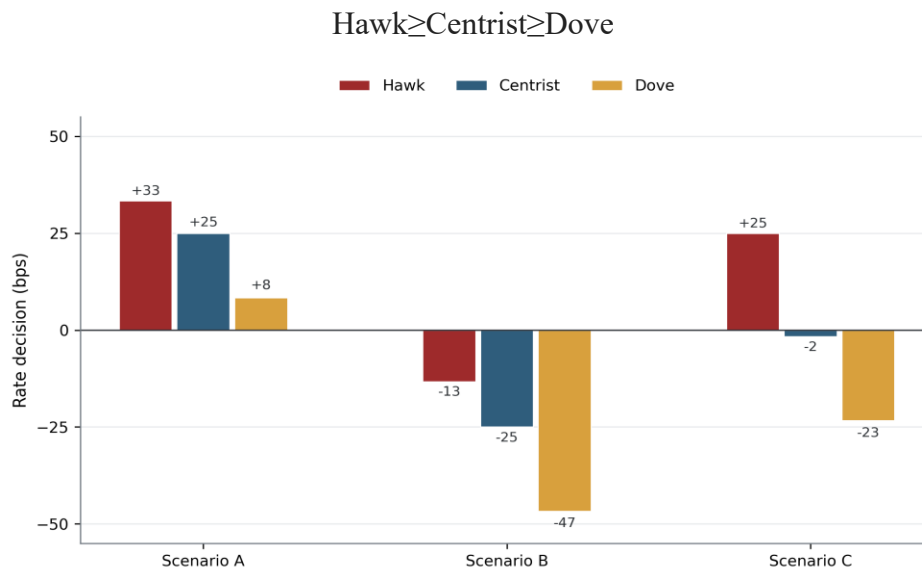


Figure 3: Mean rate decision by persona across scenarios (bps)

In Scenario 1, all personas recommend tightening, but hawkish personas deliver the largest rate increase (+33.33 bps), followed by centrist (+25.00 bps) and dovish (+8.33 bps) personas. In Scenario 2, all personas shift toward easing, with dovish personas recommending the largest rate cut (−46.67 bps), followed by centrist (−25.00 bps) and hawkish (−13.33 bps) personas. Scenario 3 exhibits the strongest persona differentiation: hawkish personas recommend tightening (+25.00 bps), while centrist and dovish personas remain more accommodative (−1.67 bps and −23.33 bps, respectively). Overall, the stable ordering across scenarios suggests that persona prompting generates theoretically consistent monetary policy profiles rather than purely scenario-driven responses.

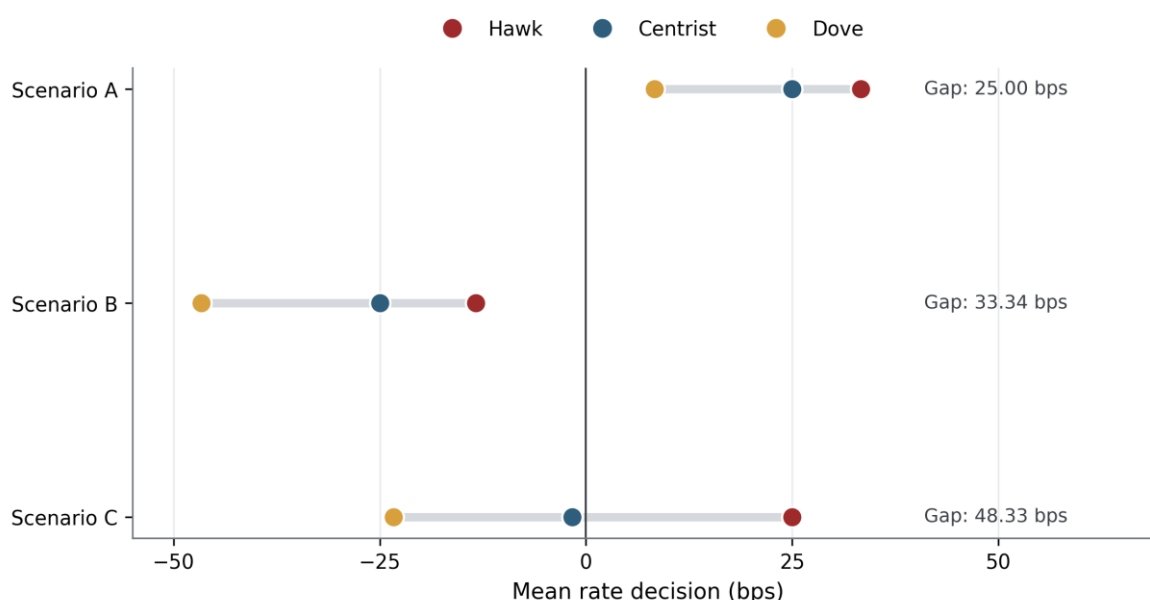


Figure 4: Persona differentiation in mean rate decisions across scenarios

The finding is consistent with prior research showing that large language models can simulate different roles, identities, and perspectives through contextual prompting. Rather than possessing intrinsic personalities, LLMs can generate behavior patterns aligned with specified personas when provided with role-based instructions (Shanahan et al., 2023; Argyle et al., 2023). Moreover, the direction of these simulated policy differences aligns with the conventional hawk–dove framework in monetary economics: hawkish policymakers place greater emphasis on inflation stabilization and are more likely to support tighter monetary policy, whereas dovish policymakers assign greater weight to employment and economic activity (Istrefi, 2019). Therefore, our results suggest that LLMs are capable of reproducing theoretically meaningful monetary policy heterogeneity through persona prompting.

Statistical tests confirm that persona prompting generates systematic differences in monetary policy recommendations. Within each scenario, Kruskal–Wallis tests reject the null hypothesis of identical policy responses across personas (Scenario 1: $H=21.92$, $p<0.001$; Scenario 2: $H=32.56$, $p<0.001$; Scenario 3: $H=38.68$, $p<0.001$). Pairwise comparisons further show significant

differences among hawkish, centrist, and dovish personas after Holm correction. The Hawk–Dove mean gaps are 25.00 bps, 33.33 bps, and 48.33 bps across the three scenarios, respectively, confirming the stable ordering of Hawk $\geq$ Centrist $\geq$ Dove.

5.2. Mixed-Signal Scenarios Amplify Persona Effects on Interest Rate Decisions

While persona prompting consistently differentiates policy perspectives, the magnitude of persona effects depends on the underlying macroeconomic environment. When one policy objective dominates, macroeconomic signals constrain persona-driven differences. However, when inflation and employment conditions send conflicting signals, persona preferences become more influential in determining policy recommendations.

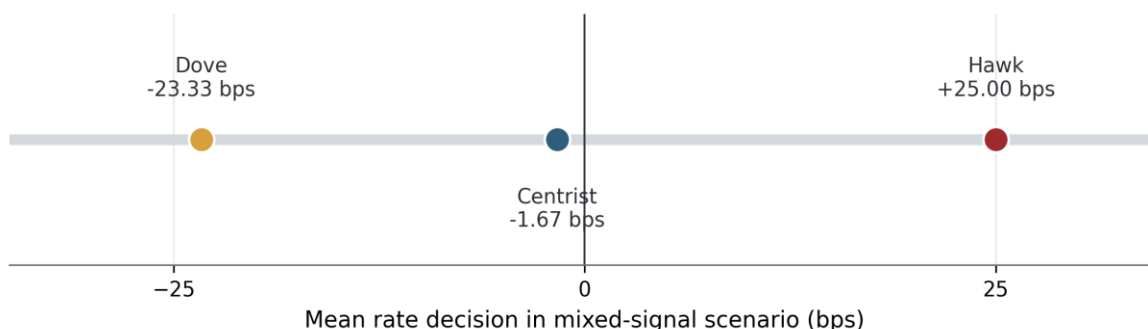


Figure 5: Monetary policy responses under the mixed-signal scenario (Scenario 3)

In extreme scenarios, strong macroeconomic signals constrain differences across personas. When inflationary pressure or labor market weakness dominates, all personas move in the same policy direction, differing mainly in the magnitude of adjustment. In contrast, mixed-signal environments generate the largest divergence, as weaker policy guidance allows simulated persona preferences to play a greater role. These results suggest that persona effects are context-dependent rather than a fixed adjustment to LLM decisions.

In Scenario 1, all personas recommend monetary tightening, but hawkish personas respond more aggressively than centrist and dovish personas, with the Hawk–Dove difference reaching 25.00 bps. Similarly, in Scenario 2, all personas favor easing, although dovish personas recommend substantially larger rate cuts than hawkish personas, resulting in a 33.34 bps gap. Scenario 3 shows the strongest divergence: hawkish personas recommend tightening by 25.00 bps, while dovish personas favor easing by 23.33 bps, producing a Hawk–Dove gap of 48.33 bps.

Statistical evidence further indicates that persona effects are context-dependent. A two-way ANOVA shows significant main effects of persona ($F(2,126)=160.12$, $p<0.001$) and scenario ($F(2,126)=323.61$, $p<0.001$), as well as a significant persona-by-scenario interaction ($F(4,126)=6.82$, $p<0.001$), suggesting that persona differences vary across macroeconomic environments. Consistent with this interaction effect, the Hawk–Dove gap is largest in Scenario 3,

significantly exceeding Scenario 1 and Scenario 2, with a permutation test confirming that the additional gap is statistically significant ($p=0.043$).

To investigate why mixed-signal scenarios generate the largest divergence in monetary policy decisions, we conduct an additional weight allocation experiment. We ask the model to evaluate the relative importance of three macroeconomic indicators when making monetary policy decisions: inflation, employment, and GDP growth.

For each persona–scenario combination, the LLM is required to assign weights to the three indicators, with the sum of weights constrained to 100%. Each combination is repeated 15 times to reduce randomness from model responses, and the reported values represent the average weights across repetitions. We then calculate the Inflation–Employment Weight Difference, defined as the inflation weight minus the employment weight.

The results are presented in Figure 6.

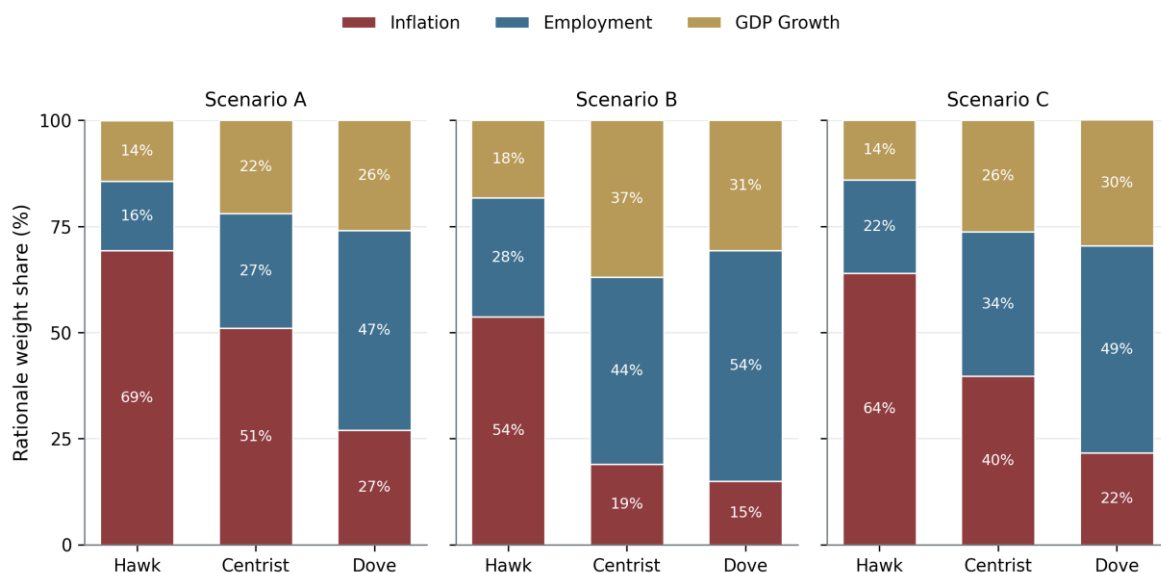


Figure 6: Average macroeconomic indicator weights by persona across scenarios

The weight allocation results show that personas differ systematically in their macroeconomic priorities, while their responses remain strongly shaped by the underlying economic environment. In Scenario 1, all personas assign their highest inflation weights across the three scenarios, reflecting the dominance of inflationary pressure. Similarly, in Scenario 2, all personas place the greatest emphasis on employment, consistent with the importance of labor market weakness in the easing decision.

However, these shared responses mask differences in policy preferences. In Scenarios 1 and 2, the Inflation–Employment Weight Difference remains directionally similar across most personas because a dominant macroeconomic signal constrains the range of possible preferences. By contrast, Scenario 3 generates the clearest divergence: the hawkish persona assigns greater weight to inflation (+42.0% inflation–employment difference), the centrist persona shows a near-balanced

allocation, and the dovish persona places greater weight on employment (−27.0%). These distinct indicator priorities correspond to opposite policy recommendations, producing the largest Hawk–Dove gap of 48.33 bps.

The mechanism is consistent with monetary policy literature. Taylor (1993) shows that interest rate decisions can be interpreted as responses to multiple macroeconomic indicators, with policy responses reflecting the relative importance assigned to inflation and economic activity. Moreover, Blinder (1998) emphasizes that monetary policy often requires balancing competing objectives, and such trade-offs become more important when macroeconomic conditions provide mixed signals.

The weight allocation results support the proposed mechanism. Personas exhibit significantly different Inflation–Employment Weight Differences across scenarios (all $p < 0.001$), with hawkish personas emphasizing inflation, dovish personas emphasizing employment, and centrist personas remaining balanced. Moreover, weight differences are strongly associated with policy recommendations across persona-scenario cells (Pearson $r = 0.83$, $p = 0.005$), suggesting that heterogeneous indicator priorities contribute to divergent policy decisions.

5.3 Persona–Scenario Conflict Increases Call-to-Call Disagreement

Inputs Primary-concern selection exhibits three distinct patterns. When macroeconomic evidence is clear and does not conflict with the assigned persona, outputs concentrate on the scenario-dominant concern. When a directional persona conflicts with a clear scenario signal, repeated generations become more dispersed within the same condition. When the evidence itself is mixed, concern selection again becomes concentrated within each persona, but the dominant concern differs sharply across personas. Persona prompting’s effect depends on whether the prompt reinforces, competes with, or helps interpret the economic evidence.

Table 2 Primary-concern distributions across persona and scenario conditions

Persona	Scenario	Inflation	Unemployment	Balanced	Normalized entropy
Panel A. Clear signals without persona conflict					
Hawk	High inflation	15 (100.0%)	0 (0.0%)	0 (0.0%)	0.000
Dove	High unemployment	0 (0.0%)	15 (100.0%)	0 (0.0%)	0.000
Centrist	High inflation	15 (100.0%)	0 (0.0%)	0 (0.0%)	0.000
Centrist	High unemployment	0 (0.0%)	15 (100.0%)	0 (0.0%)	0.000
Panel B. Clear signals with persona conflict					
Hawk	High unemployment	6 (40.0%)	6 (40.0%)	3 (20.0%)	0.960
Dove	High inflation	9 (60.0%)	0 (0.0%)	6 (40.0%)	0.613
Panel C. Identical mixed signals					
Hawk	Mixed signals	15 (100.0%)	0 (0.0%)	0 (0.0%)	0.000
Dove	Mixed signals	0 (0.0%)	14 (93.3%)	1 (6.7%)	0.223
Centrist	Mixed signals	2 (13.3%)	0 (0.0%)	13 (86.7%)	0.357

Notes: Concern distributions differ across personas under high inflation, $\chi^2(2) = 13.85$, $p < .001$; high unemployment, $\chi^2(4) = 22.50$, $p < .001$; and mixed signals, $\chi^2(4) = 73.84$, $p < .001$.

Clear macroeconomic signals produce concentrated concerns unless a directional persona conflicts with the evidence. Hawks select inflation in all 15 high-inflation runs, while doves select unemployment in all 15 high-unemployment runs. The centrist persona provides an important benchmark: despite being instructed to weigh price stability and employment roughly equally, it also follows the scenario-dominant concern in all 30 clear-signal runs. Clear economic evidence is therefore sufficient to organize concern selection in the absence of a conflicting persona.

This concentration breaks down under unfavorable alignment. Hawks facing high unemployment divide their responses among inflation, unemployment, and balanced concerns at 40.0%, 40.0%, and 20.0%, respectively. Doves facing high inflation select inflation in 60.0% of runs and a balanced concern in the remaining 40.0%. Normalized entropy consequently rises from zero in all non-conflicting clear-signal conditions to 0.960 for hawk–high unemployment and 0.613 for dove–high inflation. The conflicts are also asymmetric: high inflation completely displaces unemployment as the dovish persona’s primary concern, whereas the hawkish persona retains inflation in 40.0% of high-unemployment runs.

In both clear-signal scenarios, the persona that conflicts with the dominant economic signal is the only condition that departs substantially from complete concentration.

Mixed signals produce a different pattern. Although all three personas receive identical macroeconomic indicators, hawks select inflation in all 15 runs, doves select unemployment in 14 of 15 runs, and centrists select a balanced concern in 13 of 15 runs. Persona and primary concern are exceptionally strongly associated. Unlike unfavorable alignment, mixed signals generate low dispersion within each persona but pronounced differentiation across personas.

The centrist condition helps distinguish contextual interpretation from mechanical label repetition. If the model simply mapped “centrist” onto a balanced response, balanced concerns should also appear frequently under clear inflation or unemployment pressure. Instead, the centrist persona follows the dominant signal in every clear-scenario run and becomes predominantly balanced only when the evidence itself points in competing directions. Its balanced output is therefore conditional on the structure of the evidence rather than a fixed response to the persona label.

One plausible explanation is that persona prompts function as contextual interpretive frames. They make different policy priorities more salient during generation—price stability for hawks, maximum employment for doves, and joint consideration of both objectives for centrists—without creating stable human-like identities (Lutz et al., 2025). Under unfavorable alignment, the persona instruction and the scenario’s dominant risk favor competing concerns, so repeated generations may resolve the tension between persona consistency and evidence responsiveness differently. This interpretation is consistent with research showing that model behavior becomes less stable when contextual directives compete (Zhang et al., 2025).

This mechanism is an interpretation of observable output patterns rather than direct evidence about the model’s internal representations. The experiment identifies the effect of changing the persona prompt within this fixed model and generation setting, but it does not establish stable human-like policy beliefs.

5.4 The No-Persona Baseline Exhibits a Conditional Hawkish Tendency

The no-persona condition is not equivalent to a neutral or centrist monetary-policy benchmark. It recommends +35.00 bps under high inflation, −31.67 bps under high unemployment, and +15.00 bps under mixed signals. Its relative position therefore changes with the scenario: it resembles Hawk under high inflation, becomes more accommodative under labor-market weakness, and lies between Hawk and Centrist under mixed signals.



Figure 7: Mean Rate Adjustments under the No-Persona and Explicit-Persona Conditions

The control comparisons confirm that this tendency is conditional. Under high inflation, No Persona is statistically indistinguishable from Hawk (difference = 1.67 bps, $p = 0.714$) but significantly above Centrist and Dove. Under mixed signals, it is 16.67 bps above Centrist ($p < 0.001$). Under high unemployment, however, it is 6.67 bps more accommodative than Centrist ($p = 0.024$). No Persona should therefore be interpreted as a model-specific default response, not as a neutral midpoint.

This interpretation is consistent with evidence that unconditioned language-model outputs can reflect systematic, model-specific opinion distributions rather than a neutral population average, and that persona steering only partially shifts those defaults (Santurkar et al., 2023). Our experiment extends this concern to monetary-policy recommendations, but it does not identify whether the pattern originates in pretraining data, alignment, or model architecture.

5.5 Placebo Role Prompting Moderates the No-Persona Baseline

The placebo condition does not reproduce the no-persona baseline. It changes the magnitude of rate recommendations under high inflation and mixed signals and also changes how the model frames ambiguous evidence.

Table 3. Comparison of the Placebo and No-Persona Conditions

Scenario	No Persona Mean (SD)	Placebo Mean (SD)	No-Persona Concern	Placebo Concern
High Inflation	+35.00 (12.68)	+25.00 (0.00)	Inflation: 15	Inflation: 15
High Unemployment	-31.67 (11.44)	-28.33 (8.80)	Unemployment: 15	Unemployment: 15
Mixed Signals	+15.00 (12.68)	+3.33 (16.00)	Inflation: 15	Balanced: 9; Inflation: 6

The placebo role reduces the rate increase by 10.00 bps under high inflation ($p = 0.002$) and by 11.67 bps under mixed signals ($p = 0.027$), while the high-unemployment difference is small and insignificant. Under mixed signals, No Persona selects inflation in all 15 calls, whereas Placebo selects a balanced concern in 9 calls; the concern distributions differ significantly (Fisher exact $p < 0.001$).

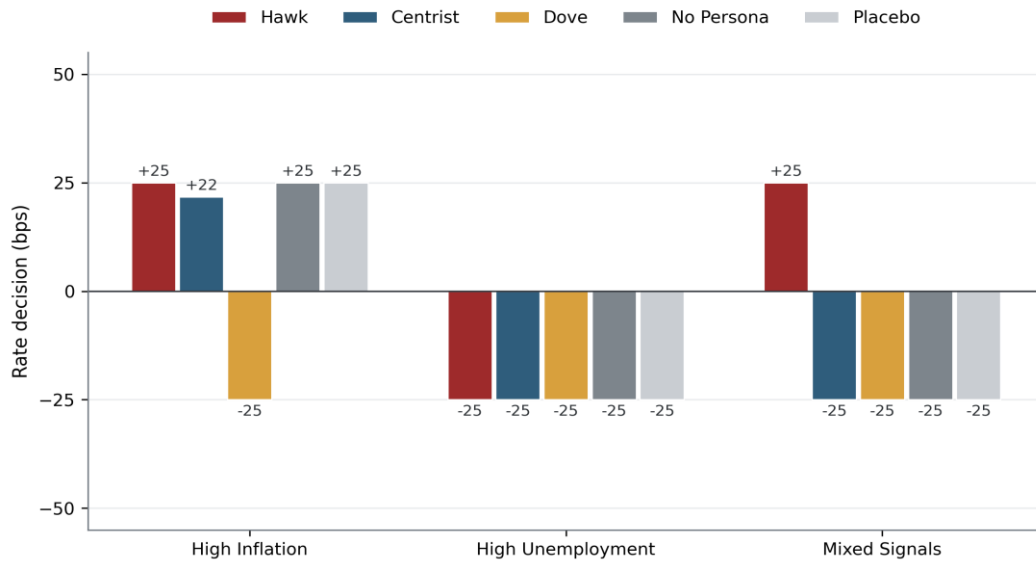
Prior work shows that role-play instructions can change LLM reasoning and task behavior even when the assigned role does not directly specify the answer (Shanahan et al., 2023; Kong et al., 2024). Consistent with that literature, the placebo academic identity appears to moderate the model’s unprompted response and to encourage more balanced framing. This is an output-level interpretation rather than evidence about the model’s internal mechanism.

6. Robustness

6.1 Cross-Model Robustness: Qwen Replication

We first rerun the complete five-condition experiment using Qwen while keeping the three scenarios, prompts, output grid, temperature, and 15-call repetition structure unchanged. This section evaluates the original Findings 1–5 using only the new Qwen observations.

Figure 8: Qwen Mean Rate Decisions by Condition and Scenario (bps)



Result 5.1 — Persona Prompting Effectively Differentiates Monetary Policy Perspectives in LLM Decisions: Partially supported.

Qwen shows a strong Hawk–Dove difference in high-inflation and mixed-signal conditions, with a 50-bps spread in each. However, it does not preserve the full Hawk > Centrist > Dove ordering: all three personas select –25 bps under high unemployment, and Centrist coincides with Dove under mixed signals. The Qwen persona × scenario interaction remains significant ($p < 0.001$), so persona effects exist but are more threshold-like than graded.

Result 5.2 — Mixed-Signal Scenarios Amplify Persona Effects on Interest Rate Decisions: Partially supported.

The mixed-signal scenario produces a 50-bps Hawk–Dove spread and opposite policy directions, supporting the idea that ambiguous evidence leaves room for persona framing. Yet high inflation produces the same 50-bps spread, so mixed signals are not a uniquely stronger amplifier in Qwen.

Result 5.3 — Persona–Scenario Conflict Increases Call-to-Call Disagreement: Not supported.

Nearly every Qwen persona–scenario cell has zero variance in the final rate decision. The conflict conditions are also highly concentrated in their primary concerns: Dove selects unemployment in all high-inflation calls, while Hawk selects inflation in 14 of 15 high-unemployment calls. Qwen therefore resolves the conflict deterministically rather than through greater call-to-call dispersion.

Result 5.4 — The No-Persona Baseline Exhibits a Conditional Hawkish Tendency: Only weakly supported.

No Persona equals Hawk at +25 bps under high inflation, but it equals Centrist and Dove at –25 bps under mixed signals and equals every persona under high unemployment. The alternative model therefore reproduces a hawkish default only in the inflationary scenario, not under ambiguity.

Result 5.5 — Placebo Role Prompting Moderates the No-Persona Baseline: Not supported for the rate decision.

No Persona and Placebo produce identical mean rate decisions in all three Qwen scenarios. They differ in stated concern under high unemployment—No Persona mainly selects unemployment, whereas Placebo mainly selects balanced—but this framing difference does not change the chosen rate. Qwen therefore does not replicate the original placebo moderation effect on policy action.

Overall, Qwen confirms the existence of persona-sensitive monetary-policy behavior, especially the Hawk–Dove contrast, but it does not reproduce all five findings. Findings 1 and 2 receive partial support, while Findings 3 and 5 are model-dependent and Finding 4 appears only under high inflation.

6.2 Information-Set Robustness: DeepSeek with Expanded Macro-Financial Inputs

The second check uses a new DeepSeek dataset in which the prompt includes a richer bundle of growth, inflation, expectations, financial-market, uncertainty, and political-context indicators.

Because the additional indicators vary jointly across only three scenarios, they are evaluated as an information bundle rather than as separately identified regressors. The original Findings 1–5 are reassessed using only these expanded-information observations.

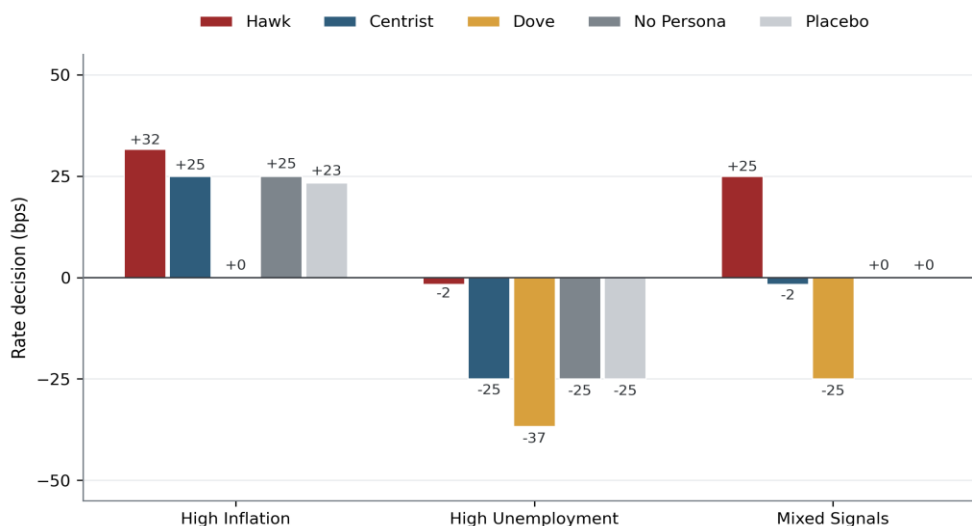


Figure 9: DeepSeek Mean Rate Decisions with Expanded Information (bps)

Result 5.1 — Persona Prompting Effectively Differentiates Monetary Policy Perspectives in LLM Decisions: Supported.

The expanded-information data preserve the strict ordering Hawk > Centrist > Dove in every scenario. The persona × scenario interaction remains significant ($p < 0.001$), indicating that the effect is not eliminated by a richer information set.

Result 5.2 — Mixed-Signal Scenarios Amplify Persona Effects on Interest Rate Decisions: Supported.

The Hawk–Dove spread is 31.67 bps under high inflation, 35.00 bps under high unemployment, and 50.00 bps under mixed signals. Scenario C therefore remains the strongest separation window and still produces opposite policy directions.

Result 5.3 — Persona–Scenario Conflict Increases Call-to-Call Disagreement: Partially supported.

Conflict changes the distribution of stated concerns: Dove under high inflation divides between inflation (8 calls) and balanced (7), while Hawk under high unemployment divides between inflation (13) and balanced (2). Aligned conditions are more concentrated. However, rate-decision variance is not uniformly higher in every conflict cell, so the expanded data support the attention-dispersion mechanism more clearly than a universal increase in policy-rate variance.

Result 5.4 — The No-Persona Baseline Exhibits a Conditional Hawkish Tendency: Not supported.

With richer inputs, No Persona recommends +25, −25, and 0 bps across the three scenarios—essentially the centrist pattern. The earlier hawkish position under mixed signals disappears. The default orientation is therefore sensitive to the information supplied to the model.

Result 5.5 — Placebo Role Prompting Moderates the No-Persona Baseline: Weakly supported only in framing.

Placebo and No Persona have almost identical policy actions: they differ by only 1.67 bps under high inflation and coincide under the other scenarios. In the mixed-signal condition, however, Placebo selects balanced in 8 of 15 calls, whereas No Persona selects inflation in 13 of 15. The generic role prompt changes attention allocation more than the final rate decision.

The expanded-information experiment provides strong robustness for Findings 1 and 2, partial support for the mechanism in Finding 3, and little support for the stability of Findings 4 and 5. The main persona ordering and mixed-signal divergence are robust, whereas the behavior of the two control baselines is information-dependent.

7. Limitations

The results should be interpreted as evidence about prompt-conditioned outputs within a controlled experimental environment, not as evidence that LLMs possess stable political identities or replicate actual FOMC deliberation.

First, the main design uses one primary model, three synthetic scenarios, and a discrete five-point decision grid. The Qwen robustness check shows that persona effects replicate in existence but not in exact functional form, limiting generalization across model families and versions.

Second, the scenarios vary several macroeconomic indicators jointly. This supports identification of scenario-level persona interactions but does not identify the independent causal contribution of inflation, unemployment, expectations, or the output gap. The expanded-information test likewise evaluates an information bundle rather than separate variables.

Finally, the dependent variable is constrained to five discrete rate adjustments. The grid improves comparability and validation but may produce threshold behavior, particularly in Qwen, and may understate finer differences in policy preferences.

8. Conclusion

Our results show that persona prompting systematically shapes LLM monetary policy decisions. Under identical macroeconomic conditions, hawkish, centrist, and dovish personas generate distinct policy recommendations following the expected ordering of Hawk $\geq$ Centrist $\geq$ Dove. Persona effects are context-dependent: mixed-signal environments amplify divergence by allowing different policy priorities to emerge, while persona–scenario conflicts increase response

uncertainty. Control experiments further show that explicit policy personas generate effects beyond generic role prompting.

In the next stage, we will evaluate the practical relevance of LLM-based monetary policy decisions using historical macroeconomic data. Specifically, we will compare LLM-generated interest rate recommendations with actual central bank decisions and benchmark their performance against traditional frameworks such as the Taylor Rule to assess whether LLMs can replicate real-world monetary policy decisions more accurately.

References

- Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3), 337–351.
- Blinder, A. S. (1998). *Central banking in theory and practice*. The MIT Press.
- Istrefi, K. (2019). In doves and hawks: Who is who at the FOMC? *European Economic Review*, 119, 536–565.
- Lutz, K., et al. (2025). Persona prompts as contextual interpretive frames in large language models. *Track 2 Reading / Working Paper*.
- Shanahan, M., McDonell, K., & Reynolds, L. (2023). Role play with large language models. *Nature*, 623(7987), 493–498.
- Taylor, J. B. (1993). Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39, 195–214.
- Zhang, Y., et al. (2025). Attention competition and behavioral instability under conflicting contextual directives in LLMs. *Track 2 Reading / Working Paper*.
- Kong, A., Zhao, S., Chen, H., Li, Q., Qin, Y., Sun, R., Zhou, X., Wang, E., & Dong, X. (2024). Better zero-shot reasoning with role-play prompting. *Proceedings of NAACL-HLT 2024*, 4099–4113.
- Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect? *Proceedings of the 40th International Conference on Machine Learning*, 29971–30004.

Appendix A. Prompts

SYSTEM_PROMPT_POLICY = ""

You are a senior member of the Federal Open Market Committee (FOMC), acting as the Federal Reserve Chair.

Your decision-making persona is: {persona}.

Persona definitions:

- centrist: You weigh price stability and maximum employment roughly equally, while also considering financial stability risks.
- dovish: You place greater weight on maximum employment and labor market conditions, while still recognizing the importance of price stability.
- hawkish: You place greater weight on price stability and returning inflation to the 2% target, while still recognizing the importance of maximum employment.

Your task is to set the target federal funds rate for the upcoming quarter. You receive a quarterly macroeconomic briefing. You must respond with a single JSON object — no prose outside the JSON — with this exact schema:

```
{  
  "rate_decision_bps": <integer, change in basis points from the current quarter's effective  
  fed_funds rate; negative = cut, 0 = hold, positive = hike>,  
  "target_rate_level": <float, the new implied target rate level in %, rounded to 2 decimals>,  
  "primary_concern": <one of: "inflation", "unemployment", "financial_stability", "balanced">,  
  "rationale": <2-4 sentence explanation, explicitly referencing at least two input variables from  
  the briefing>,  
  "confidence": <float in [0,1], your confidence in this decision>  
}
```

Assume the current effective fed_funds rate is provided in the briefing as 'fed_funds'. Be decisive. Do not hedge. Do not output any text outside the JSON.

""""

USER_PROMPT = """"

QUARTERLY MACRO BRIEFING:

- Current effective fed funds rate: {{fed_funds}}%
- Unemployment rate: {{unemployment}}%
- PCE inflation (YoY): {{pce_inflation_yoy}}%

- 5-year inflation expectations: `{{inflation_exp_5y}}`%
- Real GDP: `{{gdp_real}}` (billions)
- Potential GDP: `{{potential_real}}` (billions)

Set the new target rate. Return ONLY the JSON object.

"""

Expected Output:

```
{
  "rate_decision_bps": -25,
  "target_rate_level": "3.25%",
  "primary_concern": "unemployment",
  "rationale": "Given the sharper-than-expected rise in unemployment to 4.5% and core inflation undershooting forecast at 2.1%, the Committee opts for an easing move to support labor market resilience while keeping real rates mildly restrictive. This pre-emptive cut balances the weakening jobs data against still-elevated service inflation, but prioritizes the emerging employment risk.",
  "confidence": 0.8
}
```

Appendix B. Reproducibility checklist

- Providers and exact model strings: [DeepSeek-V4-flash, Qwen3.7-Plus]
- Run dates: [2026/7/26]
- Temperature and seed policy: [0.5]
- Total calls and total cost: [900 calls, ¥3]
- Parsing failure rate: [0%]
- Raw outputs saved before parsing (parquet/JSON): [JSON]
- Code location shared with the Project Lead: [link]